

Continual Learning with O-LoRA: Orthogonal Low-Rank Adapters for Sequential Tasks

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1. Introduction

Continual learning requires a model to sequentially acquire knowledge across multiple tasks without forgetting what it previously learned. A common challenge is **catastrophic forgetting**, where adapting to a new task overwrites representations built for earlier ones. When parameter-efficient methods such as **LoRA** are applied in this setting, the low-rank adapter matrices trained on successive tasks tend to occupy overlapping subspaces, causing destructive interference. **O-LoRA** addresses this by imposing an orthogonality constraint on the A matrices of each new task adapter with respect to all previously learned ones, ensuring each task projects inputs into a distinct region of the low-rank latent space and thereby limiting the interference that drives forgetting.

2. Dataset and Preprocessing

Models are evaluated on a fixed sequence of four text classification tasks:

- AG News** – 4-class news topic classification
- Amazon Polarity** – binary sentiment classification
- DBpedia 14** – 14-class Wikipedia topic classification
- Yahoo Answers Topics** – 10-class question topics

Each task is trained on independently, with no replay of prior task data. Samples follow the instruction tuning paradigm:

```
{task instruction}\nOption: {comma-separated label names}\n{input text}\nAnswer:
```

Only the input text is truncated when the sequence exceeds the maximum length. Cross-entropy loss is computed only on the label token; at evaluation the predicted label is the candidate label token with maximum logit at the position before **Answer**.

3. Model and Experimental Setup

- Backbone:** Qwen2.5-1.5B.
- LoRA adapters:** rank $r = 8$, scaling $\alpha = 32$, dropout 0.1, applied to the query and value projections of every attention layer.
- Optimisation:** 1 epoch / task, learning rate 10^{-3} , batch size 8.
- Compared methods:**
- IncLoRA** – interference-free upper bound: a separate adapter trained and frozen for each task (BWT = 0 by construction).
 - O-LoRA** – a single cumulative adapter penalising overlap between the new task’s A matrix and all previously learned ones via an orthogonal regularisation term weighted by $\lambda_1 \in [0.2, 0.3]$.

4. Orthogonal Regularisation

For task t with adapter $\Delta W_t = B_t A_t$, the training loss combines the classification term with a pairwise orthogonality penalty between the current A_t and every *frozen* previous A_i :

$$\mathcal{L}_t = \mathcal{L}_{\text{CE}}(A_t, B_t) + \lambda_1 \sum_{i < t} \|A_i A_t^\top\|_F^2$$

The penalty drives the row-spaces of successive A matrices into mutually orthogonal directions, so each new task projects inputs into a region of the low-rank latent space that does not overlap with prior tasks. Crucially, **no constraint is applied to the B matrices**.

5. Results

IncLoRA baseline (no forgetting by design):

Task	Accuracy
AG News	0.854
Amazon Polarity	0.956
DBpedia 14	0.972
Yahoo Answers Topics	0.640

O-LoRA – summary metrics:

Run	Samples / task	λ_1	Avg Acc	BWT
Short	500/500/500/500	0.3	0.846	−0.032
2h	500/650/1850/1300	0.2	0.799	−0.091
Full (~50%)	2100/2700/7500/5300	0.3	0.654	−0.268

Per-task accuracy: initial → final:

Task	Short	2h	Full (~50%)
AG News	0.878→0.825 (−6%)	0.842→0.747 (−11%)	0.887→0.494 (−44%)
Amazon Pol.	0.956→0.957 (~0%)	0.947→0.915 (−3%)	0.905→0.746 (−18%)
DBpedia 14	0.964→0.919 (−5%)	0.979→0.833 (−15%)	0.980→0.729 (−26%)
Yahoo Ans.	0.683→0.683 (—)	0.702→0.702 (—)	0.649→0.649 (—)

Average accuracy and BWT both **degrade as the number of training samples grows**, even though more data improves *initial* per-task accuracy. The short run is competitive with IncLoRA with very little forgetting; at ~50% of the paper’s training scale, BWT collapses to −0.268, with AG News losing over 44% of its accuracy.

6. Impact of Prompt Structure

Switching to the instruction-tuning format – which explicitly lists all candidate labels in every prompt – had a larger effect than any hyperparameter choice.

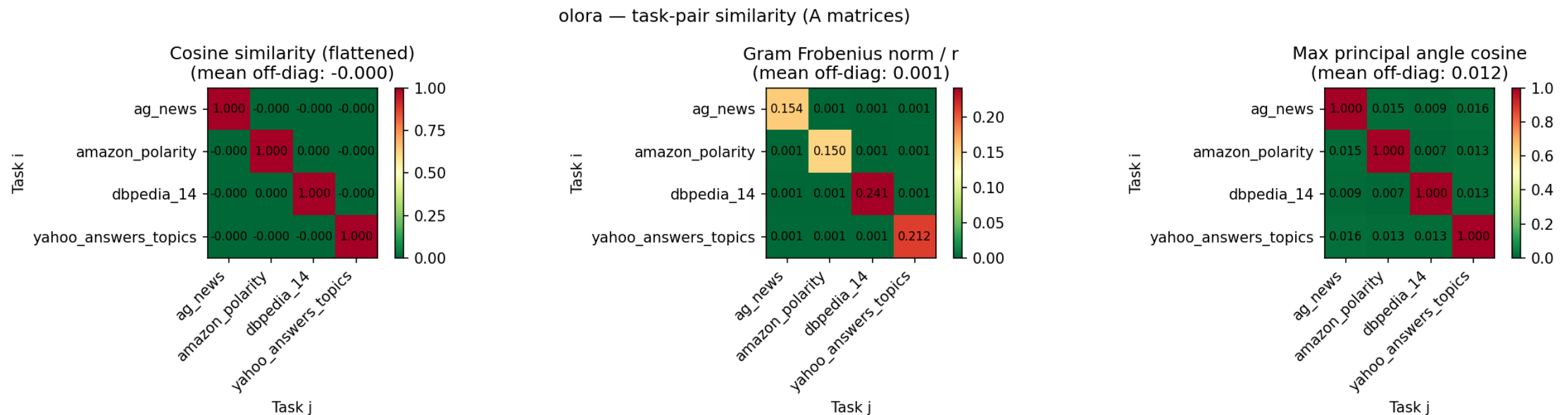
Prompt format	Avg Acc	BWT
Old (no label options)	0.597	−0.344
New (instruction + options)	0.799	−0.091

+0.20 in accuracy and +0.25 in BWT, with substantially less per-task forgetting. Listing the candidate labels in every prompt acts as a **task-specific anchor** – even as adapter weights shift, the model can recover the correct output distribution by attending to the label tokens explicitly present in context, rather than relying purely on the adapted weights to recall which labels are valid.

7. Orthogonality Analysis

Three pairwise metrics across task adapter matrices:

- Cosine similarity** – flat-vector directional alignment; blind to subspace structure.
- Gram Frobenius norm** $/r - \|A_i A_j^\top\|_F / r$; the quantity directly minimised by O-LoRA’s loss term.
- Max principal angle cosine** – largest singular value of the normalised cross-Gram between orthonormalised bases.



Pairwise similarity of A matrices across tasks – near-zero across all metrics.

All three metrics show **near-zero pairwise similarity** across tasks for A , even with $\lambda_1 = 0.2$ –0.3 (lower than the original paper). The constraint is not difficult to satisfy. While B matrices are nearly orthogonal as flat vectors (cosine ~ 0), their **column subspaces share $\sim 30\%$ of their directions** (principal-angle cosine ~ 0.29), and their magnitudes grow proportionally with training steps. Since O-LoRA imposes no regularisation on B , this unconstrained growth is plausibly the primary driver of forgetting at higher scales.

8. Conclusions

- O-LoRA’s regulariser keeps A matrices **geometrically separated** across all three metrics, with mild penalties.
- Translating geometric separation into reduced forgetting is **highly sensitive to data scale**: near-zero BWT (−0.032) at 500 samples/task, sharp collapse (−0.268) at ~50% of the paper’s scale.
- The unregularised B matrices grow in magnitude and share $\sim 30\%$ of column-space directions across tasks – a likely primary driver of forgetting at scale.
- Prompt structure matters more than hyperparameters**: explicitly listing label names yields +0.20 accuracy and +0.25 BWT.
- Non-monotonic recovery of AG News across the 8h run is consistent with unconstrained B matrices coincidentally cancelling each other’s interference at intermediate steps, rather than structured preservation of past knowledge.

9. References

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